Clustering E-commerce Product Offers

K-means · Ward agglomerative · DBSCAN · Gaussian mixture

ONE-PAGE SUMMARY
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35,311
product offers across 10 categories

25,311
unseen offers reserved for the final test

0.62
ARI of K-means on unseen offers

0.82
stability of K-means under resampling

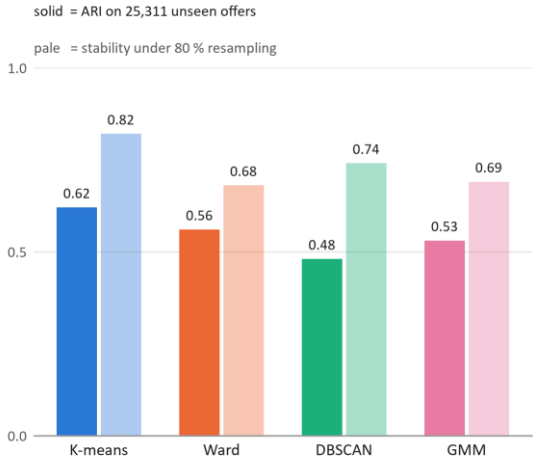
1 DATA
PriceRunner offers; 10,000 for tuning, 25,311 held out

2 FEATURES
TF-IDF, then 20-d LSA at unit length (cosine geometry)

3 MODELS
Four algorithms; k = 10 from the retailer's taxonomy

4 EVALUATION
Internal indices, ARI / NMI, stability, holdout, scalability

RESULTS AT A GLANCE



RECOMMENDATION
Deploy K-means (k = 10, normalised LSA). Send offers with confidence margin below 0.2 to review, retrain monthly, and keep DBSCAN as an outlier detector.

KEY FINDINGS

- **K-means is the best choice.** Highest accuracy (ARI 0.62, NMI 0.70), greatest stability and lowest cost, with an exact rule for scoring new offers.
- **Q1, why K-means fails.** It assumes equal-size, equal-spread, convex clusters. The predicted tipping points (size ratio 25, elongation 0.63) held in simulation: GMM and DBSCAN scored 1.0 where K-means scored about 0.
- **Q2, scaling matters.** Unscaled descriptors took 93 % of the variance and K-means grouped offers by title length (ARI 0.01). Z-scoring restored ARI to 0.54; features already on one scale gained nothing.
- **Q3, DBSCAN without guessing.** eps and min_samples set a single density level. The k-distance knee only bounds eps; a label-free grid score chose min_samples 160 and eps 0.475, the cell an oracle using labels would pick.
- **Where it errs.** Fridges, freezers and fridge-freezers overlap in wording, and German-language listings cluster by language (72 % of one cluster).

K-MEANS
Primary model: accurate, stable, cheap and directly scorable.

WARD
Small catalogues that need a hierarchy; memory grows with n squared.

DBSCAN
Outlier and near-duplicate detector; leaves 19.5 % of offers unlabelled.

GMM
When soft memberships are needed; slower and initialisation-sensitive.

LIMITATIONS

- **Titles only.** No images, attributes or multilingual embeddings. ARI 0.62 is good, not high; with labels available, a supervised classifier is stronger.
- **One dataset, one split.** K-means varies about 0.04 ARI across seeds, so its lead over Ward (+0.06) is modest.
- **Outside knowledge.** k = 10 and the 20-d embedding used it (disclosed); other tuning was label-free. Ward and DBSCAN are scored on unseen offers with a nearest-neighbour rule.
- **Timings are machine-specific.** The lasting point is Ward's quadratic growth: about 5 GB at 35,311 offers.

Full report: Clustering_Assignment_Report.docx | Code and outputs: Clustering_Assignment.ipynb